

Euclid's Elements — Book I

Proposition 43

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

In any parallelogram the complements of the parallelograms about the diameter are equal to one another.

1 Introduction

Proposition I.43 returns from the construction problem of I.42 to a purely relational statement about an already given parallelogram. A diagonal of the large parallelogram is divided at a point K . Two smaller parallelograms are placed “about the diameter”, one on each part of the diagonal. The two remaining quadrilateral regions are the classical *complements*, and Euclid proves that they are equal.

The formal reconstruction keeps the argument entirely synthetic. It does not introduce numerical area, scalar multiplication, a content order, or a cancellation axiom. Instead, it turns Euclid's Common Notion 3 step into a direct witness for the already defined relation `HilbertScissorsEquicomplementable`.

This makes I.43 a particularly clean test of the scissors language introduced at I.35. The proof is short enough that the historical and formal dependency graphs remain visibly related, but the formal representation makes explicit which triangular pieces are removed and which remain.

2 The Classical Configuration

Let $ABCD$ be a parallelogram with diagonal AC . Let K lie strictly on the diagonal, and choose points

$$A - G - B, \quad A - E - D, \quad B - F - C, \quad D - H - C,$$

so that $AGKE$ and $KFCH$ are parallelograms. The two quadrilateral regions left over are

$$GBFK \quad \text{and} \quad EKHD.$$

These are the two complements in the statement of I.43.

The Lean theorem records the strict order data explicitly:

$$A - K - C, \quad A - G - B, \quad A - E - D, \quad B - F - C, \quad D - H - C.$$

The two small parallelograms are also hypotheses rather than constructions. Thus I.43 is not a construction theorem in the current development; it is a content theorem for a given configuration.

Figure 1 shows exactly this formal configuration. The dashed segments BK and KD are not extra hypotheses. They are the diagonals used to represent the two complement regions as sums of two triangles.

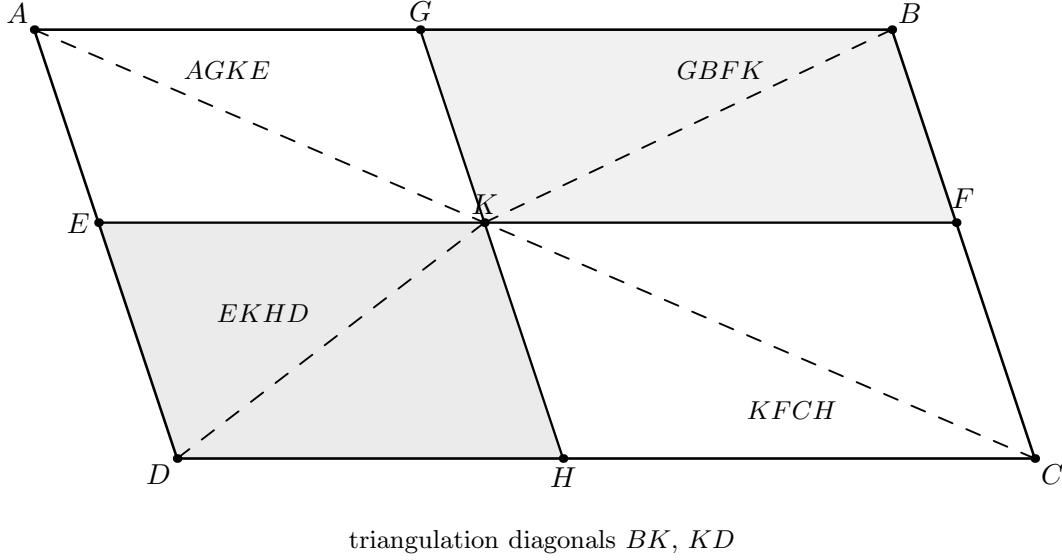


Figure 1: The configuration of Proposition I.43. The large parallelogram is $ABCD$, with $A - K - C$. The parallelograms $AGKE$ and $KFCH$ lie about the two parts of the diagonal. The remaining regions are the complements $GBFK$ and $EKHD$. In the Lean representation the first complement is triangulated by BK as $GBK + KBF$, while the second is triangulated by KD as $EDK + KDH$.

3 Classical Proof

Euclid's proof uses Proposition I.34 three times. Since the diagonal of a parallelogram bisects it into equal triangles,

$$\triangle ABC = \triangle ACD.$$

The same theorem applied to the two smaller parallelograms gives

$$\triangle AGK = \triangle AEK, \quad \triangle KFC = \triangle KHC.$$

Equivalently, in Euclid's usual lettering, these are the two equal pairs of triangles belonging to the parallelograms about the diameter.

Common Notion 2 combines the two small equalities. The corresponding pairs of corner triangles therefore have equal total content. Since the two large diagonal triangles are equal, Common Notion 3 removes equal content from equal content and leaves the two complements equal.

The classical dependency graph is therefore

$$\begin{array}{c}
ABCD, AGKE, KFCH \text{ parallelograms} \\
\Downarrow \text{ I.34 three times} \\
ABC = ACD, \quad AGK = AEK, \quad KFC = KHC \\
\Downarrow \text{ C.N.2} \\
AGK + KFC = AEK + KHC \\
\Downarrow \text{ C.N.3} \\
GBFK = EKHD.
\end{array}$$

The historical proof is therefore a forward content argument. No equal-content hypothesis is used to recover a metric, incidence, or parallelism conclusion. This places I.43 on the opposite side of the logical boundary exposed by I.39–I.40.

4 Complements and Equicomplementability

The central formal point is that Euclid’s phrase “subtract equals from equals” need not be encoded as a general cancellation rule

$$P + R \sim Q + R \implies P \sim Q.$$

No such theorem is required here.

The existing scissors relation defines formal equicomplementability by the existence of two equidecomposable complements R, S such that

$$R \sim S, \quad P + R \sim Q + S.$$

Thus I.43 can be proved by exhibiting the pieces removed by Euclid as the witnesses R and S .

For the two complement terms put

$$\begin{aligned}
P &= GBK + KBF, \\
Q &= EDK + KDH,
\end{aligned}$$

and choose

$$\begin{aligned}
R &= AGK + KFC, \\
S &= AEK + KHC.
\end{aligned}$$

Proposition I.34 gives $R \sim S$. The two full-triangle decompositions and the large diagonal equality give

$$P + R \sim Q + S.$$

These two facts are already exactly the data required by `HilbertScissorsEquicomplementable`.

This is the key difference between an abstract cancellation theorem and the present proof. The formalization does not infer equality after deleting an unrepresented common summand. It records the deleted pieces explicitly as part of the certificate.

5 Proof Architecture of the Reconstruction

The working theorem `i43_raw_complements` has five mathematical stages.

5.1 Step 1: decompose the first large diagonal triangle

Using $A - K - C$, the triangle ABC is split at K . The resulting AK -piece is split at G using $A - G - B$, and the KC -piece is split at F using $B - F - C$. The result is

$$ABC \sim (AGK + GBK) + (KBF + KFC).$$

In the Lean file this is the local identity `hT1`.

5.2 Step 2: decompose the second large diagonal triangle

The mirror construction uses $A - E - D$ and $D - H - C$. After orienting the first betweenness relation as $D - E - A$, the triangle ACD is decomposed as

$$ACD \sim (EDK + AEK) + (KDH + KHC).$$

This is `hT2`.

5.3 Step 3: obtain the three I.34 equalities

The big parallelogram gives

$$ABC \sim ACD.$$

The two parallelograms about the diagonal give

$$AGK \sim AEK, \quad KFC \sim KHC.$$

These are the local facts `hT3`, `hT4`, and `hT5`. The implementation obtains all three through the already proved theorem `euclid_proposition_34_diagonal` and the congruent-triangle generator of the scissors calculus.

5.4 Step 4: assemble and regroup the full decompositions

Transitivity gives one four-piece relation

$$\begin{aligned} & (AGK + GBK) + (KBF + KFC) \\ & \sim (EDK + AEK) + (KDH + KHC). \end{aligned}$$

The code then uses only associativity and commutativity of multiset addition to regroup this as

$$(GBK + KBF) + (AGK + KFC) \sim (EDK + KDH) + (AEK + KHC).$$

No geometric theorem is hidden in this regrouping.

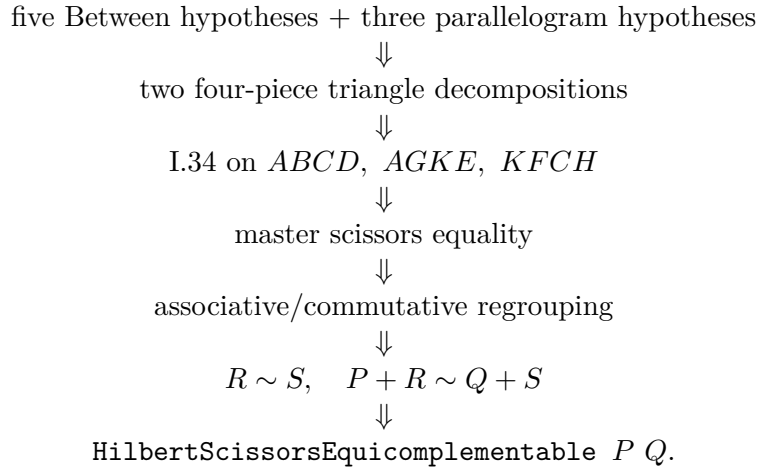
5.5 Step 5: package the equicomplementability witness

Finally `hT4` and `hT5` are combined with `HilbertScissorsEq.add`, giving

$$AGK + KFC \sim AEK + KHC.$$

Together with the regrouped master relation, these are exactly the two fields required by the definition of equicomplementability.

The complete Lean DAG is therefore



6 Lean Formalization

The file imports only the already established diagonal theorem for parallelograms and the scissors calculus:

```
import CGJteamLab.Proposition34
import CGJteamLab.HilbertScissors
```

The substantial proof is isolated in the proposition-local helper `i43_raw_complements`. Its public shape is the following.

```
theorem i43_raw_complements
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D K G E F H : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D)
  (hAKC : Geo.Between A K C)
  (hAGB : Geo.Between A G B)
  (hAED : Geo.Between A E D)
  (hBFC : Geo.Between B F C)
  (hDHC : Geo.Between D H C)
  (hPar1 : IsParallelogram Geo A G K E)
  (hPar2 : IsParallelogram Geo K F C H) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo G B K +
     hilbertScissorsTriangle Geo K B F)
    (hilbertScissorsTriangle Geo E D K +
     hilbertScissorsTriangle Geo K D H) := by
  -- proof in Proposition43.lean
```

The final Book I endpoint is deliberately thin:

```
theorem euclid_proposition_43
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
```

```

(A B C D K G E F H : Geo.Point)
(hParallelogram : IsParallelogram Geo A B C D)
(hAKC : Geo.Between A K C)
(hAGB : Geo.Between A G B)
(hAED : Geo.Between A E D)
(hBFC : Geo.Between B F C)
(hDHC : Geo.Between D H C)
(hPar1 : IsParallelogram Geo A G K E)
(hPar2 : IsParallelogram Geo K F C H) :
HilbertScissorsEquicomplementable Geo
  (hilbertScissorsTriangle Geo G B K +
   hilbertScissorsTriangle Geo K B F)
  (hilbertScissorsTriangle Geo E D K +
   hilbertScissorsTriangle Geo K D H) := by
exact
i43_raw_complements
  Geo
  A B C D K G E F H
  hParallelogram
  hAKC
  hAGB
  hAED
  hBFC
  hDHC
  hPar1
  hPar2

```

The final certificate returned by the raw helper uses

```

hilbertScissorsTriangle Geo A G K +
hilbertScissorsTriangle Geo K F C

```

and

```

hilbertScissorsTriangle Geo A E K +
hilbertScissorsTriangle Geo K H C

```

as the two equidecomposable complements.

7 Relationship with Earlier Propositions

7.1 Proposition I.34

I.34 is the only earlier Book I proposition used directly by I.43. It enters three times through `euclid_proposition_34_diagonal`: once for the large parallelogram and once for each parallelogram about the diagonal. The current proof does not replay the parallel-angle or congruence argument behind I.34; it consumes the already established diagonal triangle congruence.

7.2 Propositions I.35–I.41

These propositions established and exercised the synthetic content language, but I.43 does not call them by proposition name. Their architectural legacy is **HilbertScissors**: finite triangle sums, splitting, congruent replacement, addition, transitivity, and equicomplementability.

In particular, I.43 is not a new converse theorem of the I.39 type. It needs no proper-part contradiction and no positivity principle.

7.3 Proposition I.42

I.42 is a construction theorem: it creates a parallelogram with prescribed content and angle. I.43 is complementary in character. The entire geometric configuration is supplied as input, and the theorem proves a relation between the two remaining regions.

7.4 Proposition I.44

The next proposition uses I.43 immediately. Historically, I.44 first creates a parallelogram of the required content and angle and then uses the complement theorem to transfer that content to a parallelogram applied to the prescribed segment. Thus I.43 is not an isolated scissors identity; it is the bridge from I.42 to the application-of-areas construction in I.44.

8 Hilbert and Book Zero Components Used

8.1 Betweenness and incidence

The five strict betweenness hypotheses are the hidden configuration data needed for the formal triangle splits. They guarantee that K, G, E, F, H lie on the intended segments, not merely on the corresponding carrier lines. Apart from reversing $A - E - D$ to $D - E - A$ for one split, no new order theorem is introduced.

8.2 Parallelogram diagonal congruence

The metric input is entirely packaged by I.34. The current proof therefore uses the developed Euclidean parallelogram layer rather than reconstructing the three congruence proofs locally.

8.3 Scissors content infrastructure

The proof uses only existing operations:

```
HilbertScissorsEq.split
HilbertScissorsEq.congruent
HilbertScissorsEq.add
HilbertScissorsEq.symm
HilbertScissorsEq.trans
scissors_congruent
scissors_triangle_swap12
scissors_triangle_swap23
```

Together with associativity and commutativity of multiset addition, these are sufficient to reconstruct Common Notions 2 and 3 in the present configuration.

No new theorem is added to `HilbertInterface`, `HilbertBookZero`, or `HilbertScissors`.

9 Source Audit

9.1 Euclid, Heath, and Joyce

The classical sources agree on the short proof: I.34 gives equality of the two large diagonal triangles and of the two pairs of small diagonal triangles; C.N.2 combines the small equalities; C.N.3 leaves the two complements equal. Joyce also emphasizes the operational meaning of “parallelograms about the diameter” and notes that the next proposition uses I.43 to change the shape of a parallelogram without changing its content.

9.2 Wilkins

Wilkins is used here as a control on the classical dependency graph and the configuration around the diagonal. No additional hidden case split is needed in the current Lean theorem because all five segment-order relations are already explicit hypotheses.

9.3 Hilbert

The closest Hilbertian source is Chapter IV, Section 18. Immediately before Hilbert’s Theorem 43, the general theory states that if equidecomposable polygons are removed from equidecomposable polygons, the remaining polygons are equicomplementable. Hilbert’s Theorem 43 then supplies the corresponding transitivity-style properties for equidecomposability and equicomplementability.

The numerical coincidence is worth avoiding: Hilbert’s Theorem 43 is not Euclid’s Proposition I.43. The connection is structural, not numerical. The present Lean proof specializes the general removal pattern directly at the term level by exhibiting the removed pieces as the two complements in the definition of `HilbertScissorsEquicomplementable`.

9.4 Hartshorne

Hartshorne’s area theory is a semantic control on the direction of the argument. I.41–I.45 belong to the forward equal-content part of Book I; the extra strictness property used by the converse propositions I.39–I.40 is not needed. This supports the present choice to remain entirely inside synthetic cut-and-paste content rather than numerical area.

9.5 Beeson, Borsuk–Szmielw, and Greenberg

These sources add no separate I.43-specific theorem to the current architecture. Beeson remains a control on omitted formal conditions, while Borsuk–Szmielw and Greenberg control the

underlying Euclidean parallel and parallelogram infrastructure. The actual proof is already expressed more directly by I.34 plus the existing scissors calculus.

10 Formalization Notes

10.1 Geometry

The formal theorem assumes more explicit positional data than Euclid writes in the proposition statement:

- K is strictly between A and C ;
- G is strictly between A and B ;
- E is strictly between A and D ;
- F is strictly between B and C ;
- H is strictly between D and C ;
- the large and two small quadrilaterals are explicitly `IsParallelogram` hypotheses.

These hypotheses are not auxiliary constructions introduced by the proof. They are the formal unpacking of the classical diagram required by the splitting constructor.

10.2 Content

The content argument is entirely forward. There is no appeal to `HilbertScissorsPositivity`, no whole-greater-than-part theorem, and no cancellation principle. The two deleted collections of corner triangles are retained as explicit witnesses of equicomplementability.

This is exactly why I.43 remains inexpensive after I.39: Common Notion 3 is not being used to infer geometry from equality of content. It is being realized as a constructive scissors certificate.

10.3 Representation

The public theorem does not return the two complements as `hilbertParallelogramTerm`. Instead it returns the exact triangulated terms used by the proof:

$$GBK + KBF \quad \text{and} \quad EDK + KDH.$$

This is faithful to the current term-level semantics of `HilbertScissors`, which represents finite sums of triangles rather than polygonal interiors. The theorem therefore establishes the equal-content relation needed for I.43 without claiming a stronger polygon-semantics result that the library does not yet encode.

No nonoverlap predicate, rectilinear-region type, or numerical area function is introduced.

10.4 Axiomatic status

The public theorem is stated under

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

because its I.34 dependency belongs to the developed Euclidean parallelogram layer. No continuity axiom is used locally, no Archimedean principle is used, and I.43 adds no new geometric or content axiom.

11 Dependency Summary

The actual formal dependency chain is

```
Hilbert incidence, order, congruence, and Euclidean parallel axioms
      ↓
developed parallelogram infrastructure
      ↓
Proposition I.34 diagonal triangle congruence
      ↓
HilbertScissors split/congruent/add/trans operations
      ↓
hT1, hT2, hT3, hT4, hT5
      ↓
master four-piece scissors equality
      ↓
explicit equicomplementability witnesses
      ↓
i43_raw_complements
      ↓
euclid_proposition_43.
```

New geometric axiom:	no
New content axiom:	no
New proposition-local helper:	yes — <code>i43_raw_complements</code>
New reusable helper:	no
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
Numerical area introduced:	no
Polygon semantics introduced:	no
Full lake build:	clean
sorry in <code>Proposition43.lean</code> :	no

12 Conclusion

Proposition I.43 is a compact confirmation that the scissors architecture built from I.35 onward has reached the right level of abstraction. Euclid’s proof uses I.34 together with addition and

subtraction of equal figures. The Lean proof preserves the same mathematical structure without numerical area and without introducing a general cancellation theorem.

The decisive formal move is to keep the pieces that Euclid “subtracts” as explicit data. The two collections of corner triangles are proved scissors-equivalent, and the two complements become equidecomposable after those collections are adjoined. That is precisely the existing definition of equicomplementability.

Thus I.43 introduces no new foundational debt. Its one local helper packages a configuration-specific decomposition, while the reusable mathematics remains where it already belongs: I.34 for the diagonal congruences and **HilbertScissors** for finite cutting, congruent replacement, addition, and transitivity. This also leaves the project in the natural position for I.44, where the complement theorem becomes an active construction tool.